The Chinese University of Hong Kong
Department of Computer Science and Engineering
Formal Languages and Automata Theory (CSCI3130)
Time: Nov. 2, 2026 – Nov. 9, 2026

Homework 4

Q1. (20%) Remove all unit-productions, all useless productions, and all λ-productions from the grammar:

$$\begin{aligned} S &\to aaA \mid aBB, \\ A &\to aA \mid \lambda, \\ B &\to bB \mid bbC, \\ C &\to B. \end{aligned}$$

What language does this grammar generate?

Q2. (30%) Convert the following context-free grammar into Chomsky normal form (CNF). Show the intermediate steps.

$$\begin{aligned} S &\to AB \mid a, \\ A &\to aAB \mid b, \\ B &\to bB \mid bAC, \\ C &\to c. \end{aligned}$$

Q3. (20%) For each of the following PDAs, determine whether it is deterministic. Justify your answers. All unlisted transitions are empty.

1. $$\begin{aligned} \delta(q_0, a, z) &= \{(q_0, Az)\}, \\ \delta(q_0, \lambda, A) &= \{(q_1, \lambda)\}. \end{aligned}$$

2. $$\begin{aligned} \delta(q_0, a, z) &= \{(q_0, Az)\}, \\ \delta(q_0, \lambda, z) &= \{(q_1, z)\}. \end{aligned}$$

Q4. (30%) Construct a nondeterministic pushdown automaton (NPDA) that accepts the following language:

$$L = \{a^n c b^n : n \geq 0\}.$$

Draw its state transition diagram. Use acceptance by final state.